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The elements of bodies, earth and society

GO ROUND A HOSPITAL LABORATORY and count the ways to examine the elements of bodies – by post-mortem anatomies or microscopical specimens, by chemical analysis of blood and solid tissues, by immunological testing for antibodies, by electrical sensors such as electrocardiograms (ECGs) and electroencephalograms (EEGs), or by imaging techniques from X-rays to magnetic-resonance scanners. For 200 years we have been finding new ways to analyse bodies (Amsterdamska and Hiddinga, 2000).¹

This chapter explores the genealogies of these forms of analysis, and the related analyses of plant and animal structures, processes and forms. It also includes earth sciences, and the social sciences which were invented *c.*1800. It complements the previous chapter and together they map the massive restructurings of STM which were effected from the late eighteenth century. At the end of this chapter I explore ways in which we might attain some *explanation* of these creations and their consequences.

Medical analysis: corpse and patient

Major public dissections were highlights of academic medicine from the Renaissance. Initially, they were meant to illustrate Greek understandings of how the body worked; they exemplified the classical analytical traditions associated with Aristotle and Galen – the humours which made up the body, the central role

of the liver in converting food to venous blood, and of the left side of the heart in transmitting the 'breath of life' from the lungs to the arterial blood. They became explorations of the bodily landscapes, a kind of natural history which revealed a landscape with many new features, including the lymphatic vessels full of watery fluid – the rivers of the body. But anatomies were also associated with a radical revision of part of the classical analysis. Just as in planetary astronomy, where the sun replaced the earth at the centre of the world, so by the demonstrations of the English physician William Harvey, the relations of the body organs were turned upside down. In Harvey's book of 1628, *On the Motion of the Heart and of the Blood*, the heart became central, pumping blood round and round the body, where previously blood had been supposed to seep gently from its source in the liver into the flesh to which it gave substance. But what was the circulation for? The lack of adequate answer was one reason for restricting the enquiry to '*matters of fact*' rather than the '*purposes*' of the Greek tradition (French, 1994; Pomata, 1996; and see Cunningham, 1997).

As in the physical sciences, this radical revision of the classical analysis prompted much investigation and, though the discovery grew from an Aristotelian tradition, the restructured organism was often presented as mechanical; René Descartes wrote an essay on the body as a mechanism, centred on the heart as a pump. But though the mechanical analogy proved fruitful in analysing the operations of the muscles and bones (Cook, 1990), the effects on medicine generally were minor. Some of the explanations changed, but the practice of medicine remained much the same; its form remained biographical and natural historical.

By the mid-eighteenth century, in physiology as in chemistry and microscopy, the attempt to analyse bodies as matter in motion was running out of steam. Instead, investigators began to examine the general properties of living matter, sometimes as they appeared in the 'simplest of living forms' such as the polyp or *Hydra*, which served the eighteenth century much as the *Amoeba* served the nineteenth and early twentieth (Canguilhem, 1975; Delaporte, 1994; Jacob, 1988; Roger, 1971; Schofield,

1969). As in the physical sciences, investigators began to look for the elements of *particular domains*, rather than assuming that the first principles of *all nature* were mechanical, or the old Greek elements. In the creation of new biomedical, earth and social sciences, as for those discussed in the last chapter, the aspirations of the technical professions were crucial.

By the later eighteenth century, anatomy had become the science of surgeons. Elite surgeons, individually or collectively, were investing their wealth in the 'cultural capital' of anatomical museums, often associated with the dissecting rooms where their students practised anatomy 'hands on'. In most European countries, leading surgeons were approaching the status of physicians – not just by anatomising, but by using public hospitals for surgical operations, by proving useful to military authorities, and generally by appealing as 'practical men' to the expanding bourgeoisie. As their influence grew, especially in hospital practice, they popularised their more anatomical understandings of disease (Fissell, 1991; Lawrence, 1996). However, because surgeons usually remained subordinate to physicians, their anatomical approach still operated within the hegemony of a natural-historical, biographical medicine based on systemic disturbance, whether of humours, fibres or nerves. More generally, the control exercised by lay-governors over charity hospitals in Britain, and of religious orders over the public hospitals of France, ensured that pastoral care usually took precedence over the acquisition of knowledge (Risse, 1999).

It was in France, as a result of the Revolution, that the power-orderings were reversed, at least in a few key teaching institutions. Political change was rapid and deep, and so was institutional change in medicine (and science and technology). Old power structures, including the Paris Faculty of Medicine, were swept away, and with them the support for classical medicine of a literary and biographical form. Aspirations were released – for a medicine which *unified* physic and surgery; for a more practical, more surgical, medicine, taught by the best qualified to the most promising; and for medical control over the hospital patients, alive and dead.

In Paris, before the Revolution, the poorhouses were run by Catholic nuns; in the Hôtel Dieu, just by the cathedral of Notre Dame, thousands of sick poor were tended unto death. Some of Paris's leading surgeons were attached to the hospital, and they tried to use the inmates for the systematic teaching of their students and apprentices – but the nuns resisted the intrusion. After the Revolution, however, the power of the Church was reduced and doctors gained control over the running of the hospitals. They were able to arrange the patients according to their diseases, conduct classes on the wards, use students as assistants and conduct autopsies on the dead paupers. It was in and around these 'medicalised' poorhouses that analytical medicine was invented and tissues became the new elements of the body (Gelfand, 1980; Risse, 1999; Vess, 1974).

The inventors were teachers associated with the new School of Health (later the School of Medicine), established in 1794 by a Revolutionary government bombarded with complaints about quackery and in desperate need of doctors for a conscript army now at war with the rest of Europe. Leading surgeons, physicians and medical reformers were recruited to staff the new school and to devise a curriculum. Initially, they improvised to meet the emergency; then they worked out a programme which promoted anatomical notions of disease as the basis for the common training of both physicians and surgeons. Students and staff were selected by competitive examination and supported by the State. The opportunity was enormously exciting – a chance to lead the reform of medicine. Paris became the centre of the world for medical students and investigators (and retained that status for fifty years). In the School of Medicine, as for engineering in the post-Revolutionary military polytechnic, dreams of analysis now coincided with the resources to realise them (Gelfand, 1980; Duffin, 1998; Maulitz, 1987; Risse, 1999; Vess, 1974).

The hospitals supplied lots of bodies, and young surgeons dissected hundreds of them. The medical students frequented the Museum of Natural History and learned comparative anatomy; they knew about the recent triumphs of French chemists, especially Lavoisier; they also read philosophy, then called *ideology*

and aiming to provide a scientific account of how ideas were formed and employed. As a science of man, ideology was sometimes regarded as a branch of medicine, and it taught the importance of analysis as a way of knowing (Albury, 1972; 1977).

We know that the brightest of the young surgeons, Xavier Bichat, intended his 'tissues' as the elements of general anatomy. They were the basic structures, homogeneous to the eye, from which the human body was composed; usually twenty-two were listed, including bone, skeletal muscle, liver, etc. The idea was not altogether new: Aristotle's works included discussions of heterogeneous and homogeneous parts, and some British anatomists had focused on membranes as common to many different organs, but Bichat generalised the questions and provided systematic answers. He catalogued the tissues, exploring their 'sensibility' and their 'contractility', and their reactions to chemical reagents. He also studied the ways they became diseased (Haigh, 1984; Pickstone, 1981).

It was in *pathological* anatomy that tissues proved most fruitful for medicine, providing a framework for the view of *diseases as lesions*. Diseased corpses could be systematically surveyed to see which tissues were affected, and how. The new geography of the body proved profitable because lesions in an organ were often restricted to particular tissues, e.g. pericarditis – inflammation of the membrane around the heart. Or inflammation might affect similar tissues in various organs – sometimes because of the continuity of tissue systems, e.g. of the lymphatic vessels or the veins (Maulitz, 1987). Throughout the nineteenth century, the analysis of tissue lesions was a major part of medicine; in big city hospitals 'pathologists' came to be employed full time to dissect and to teach. From the 1830s microscopes were deployed, and soon thereafter, as we shall see below, it became possible to analyse tissues and lesions in terms of their constituent *cells*. We have already seen that the new chemistry could be applied to medicine, for the analysis of normal and diseased fluids and tissues. So now we begin to see the genealogies of the many analytical methods which I noted in the introduction to this chapter. But the corpse-analysis of the Paris teaching hospi-

tals had a further extension – to the clinical examination of the living patient.

When diseases were but disturbances of equilibrium, searching for them in the body was no more rational than searching for the site of hyperinflation in a present-day economy. But if diseases were lesions – inflammation of tissues, say – they could in principle be *found*. Maybe they could be located and assessed before death; the patient could be analysed before it became a corpse. And sure enough, the Paris hospitals which demonstrated tissue analysis to their students also took up auscultation ('sounding' the chest), and then the *stethoscope* – a wooden tube between the body of the patient and the ear of the doctor. By transmitted sounds, a doctor could 'see' the diseased lungs or heart; he was trained to correlate sounds with the pathology revealed at autopsy. Thus clinical examination began to take its familiar form. In nineteenth-century teaching hospitals diagnosis was all; therapeutics was a sideshow. Doctors in the big continental hospitals were often sceptical about customary remedies, inclined to let diseases 'run their courses' – the patterns of pathology were clearer that way.

The ramifications of analysis for medical organisation proved very extensive. In as much as diseases were specific, anatomical and *local*, doctors could legitimately specialise in particular organ systems. That was the basis of many of the nineteenth-century special hospitals – for diseases of the eye, etc. – often resisted by general physicians and general surgeons, but pushed by ambitious young men anxious to corner a part of the medical market. In Britain such medical entrepreneurs often boasted of training in Paris, where medical teaching was so diversified as to offer lots of arenas for the development and dissemination of specialist knowledges (e.g. Granshaw, 1985). And if diseases were specific and defined, you might treat them 'en bloc' in hospital wards and compare the numerical results of different treatments. The 'numerical method' of the Paris hospitals, though not entirely without precedent, is the basis of hospital statistics as developed in the twentieth century to provide measures of efficacy in medicine. Statistical analysis was built on pathological

analysis, in hospitals (and also in 'public health') (Matthews, 1995).

The medical alternatives to analysis

So far, I have told a unidirectional tale of the advance of analysis in medicine, as if the analytical medicine of (some) teaching hospitals was the whole story. Indeed, history of medicine is often presented in this way – 'hospital medicine' replaced 'bedside medicine' (and was itself replaced by laboratory medicine).² But this is a misleading travesty. Thinking about ways of knowing, and ways of making, allows a more complex, political and helpful story, for medicine and for other technologies.

As I have tried to stress in earlier chapters, natural-historical or biographical medicine was not *replaced*, even in teaching hospitals. For decades after the Revolution, many French hospitals continued to practise a medicine that was chiefly biographical. In Britain, the charity hospitals in which medical teaching was conducted remained wary of appearing as places of dissection and 'experiment', for fear of dissection after death was widespread in Britain c.1830, and the 'resurrection' of corpses was a huge scandal, even before Burke and Hare were discovered 'making corpses' for Edinburgh anatomists (Richardson, 1988). In any case, British medicine was dominated throughout the nineteenth century by physicians and surgeons whose chief pride was their lucrative private practices. They did not see themselves as medical scientists; they taught bedside skills that would prove useful when their protégés set up as general practitioners (and referred their difficult, rich, cases to their old masters). The system did not produce pathological discoveries at the same rate as the big continental teaching hospitals, but it may well have provided better doctors and better care for the urban poor.

In Victorian Britain, 'medical science' tended to be the preserve of young men, not yet established as consultants, eager to win a name as teachers, or to impress professional societies by showing their findings or by importing continental knowledge. They aspired to be 'analysts' in a world of 'experience' (of craft and

natural history). By the end of the century, a few such would-be medical scientists had gained salaried posts as anatomists, physiologists or pathologists in medical schools. They included the pioneers of bacteriology – a new form of medical analysis which was taken up by public health doctors (a characteristic linkage of technical analysis and social management).³ These ‘scientists’ in medicine fought to make medical schools more like science faculties, including full-time clinical professorships; and they dominated the Medical Research Council (MRC) founded by the government just before the First World War. They then used the MRC to push standardisation (and experimentalism) in medicine, but not until the mid-twentieth century would their way of working become dominant among elite hospital clinicians. Throughout their long campaign for a more analytical and experimental medicine, they drew on continental, and then American, examples.

In Austria and Germany, some of the big-city teaching hospitals (Vienna and then Berlin) had been strongholds of analysis from the early nineteenth century (Duffin, 1998), but most medical schools were in universities in small country towns. Had they wanted to adopt ‘mass’ pathological anatomy, they did not have the numbers of patients required for efficiency. In fact, many early nineteenth-century German physicians saw a future, not in mass dissections but in the ever closer examination of individual patients. In a culture that placed great stress on ‘development’ – in embryos, lives or societies – charting the unfolding of diseases in the biographies of patients was both intellectually fulfilling and a practical way of teaching. In the later nineteenth century, with the rise of experimental physiology, laboratory methods could be bent to these ends. Physiological medicine could be more individual, less ‘museological’, than the pathology of Paris and Vienna (Tuchman, 1993).

Doctors in the USA drew on all these European traditions. In the early nineteenth century, many Americans had studied in Paris, especially for anatomy and surgery; later medical students went mostly to German universities, to learn pathology, physiology and bacteriology. By the end of the century, the returnees had

managed to create a few full-time posts for medical scientists, even for scientific clinicians. Backed by big philanthropists such as Rockefeller and Carnegie, whose agents were much impressed by science, they created German-type departments and hospitals in a few East Coast cities. The pattern spread rapidly in the twentieth century, partly to raise the status of doctors by restricting access to medical schools; and Americans promoted it in Britain from about 1910 (Warner, 1991; 1994).

However, in all countries, in as much as doctors dealt with individual patients and helped guide their lives, biographical medicine necessarily remained part of practice. Even in the most soulless of continental teaching hospitals, case histories were taken, if only to check the results of the physical examination, and where patients were paying for treatment, no doctor could afford to be too analytical. Young American doctors were warned *not* to bring home the doctor-patient relations they saw in the Paris poorhouses; new knowledge gained from hospital autopsies could be useful, but it had to be subordinated to the consideration of the individual, paying, patient (Warner, 1985).

We see that the relations between analytical and natural-historical medicine were always complex, dynamic and context-dependent (as they were also in other kinds of STM). In the twentieth century, as analytical methods developed further, some doctors reacted by advocating a return to bedside medicine, or to a more 'holistic' approach or to the study of 'the natural history of diseases'. These movements were significant in all the major countries between the wars, and since the Second World War we have seen recurrent attempts to stress 'biographical' aspects of general practice, including psychosomatic and psychosocial dimensions.

The dialectic is inevitable and healthy (Lawrence and Weisz, 1998); in all practical activities there will be tension between the claims of formal analysis and those of 'experience' which are hard to formulate; we see such tensions in industry and agriculture as well as in medicine. One advantage of 'ways of knowing' is that such problems can be seen 'across fields', comparatively, so that we can learn from a wider range of other cases.

In the next section we return to Paris *c.* 1800, to examine analysis in the biological sciences, but bear in mind that these were often closely related to medicine, from their origins to our present.

Analysing plants and animals

The National Museum of Natural History had been the Royal Botanical Garden (and Menageries). Because natural history was well regarded by the Revolutionary government, and because those responsible for the Royal Gardens were politically nimble, the garden was not abolished with the other royal institutions of the *ancien régime*. Instead it was 'nationalised' and its staff expanded by the appointment of new specialists, who were meant to form a republic of professors, each with their own section of nature (Spary, 1995). That was not the outcome. By 1800 Georges Cuvier dominated the institution, at least on its animal side; he pioneered a structural-functionalist analysis of animals, though his colleagues championed very different approaches – as we shall see.

Cuvier's job, as a curator-professor and a rising member of the scientific associations of Paris, was to discipline his domain – to order it, but also to show that studying animals was as scientific as the new chemistry, or the mathematical physics then championed by Pierre Simon Laplace. For this purpose, zoology could no longer be a matter of arranging snail shells – Cuvier would leave this to his colleague Lamarck, who annoyed him by claiming that animal species could change into other animal species. To Cuvier such 'transformism' was nonsense, based on a superficial natural history that failed to get to grips with comparative anatomy and the deep-seated, structural differences between animals. He came to argue that the animal kingdom comprised four chief groups, each with a defined basic plan – radiates (jellyfish, sea urchins, etc.), molluscs, segmented invertebrates (e.g. worms and insects) and vertebrate animals. To understand how animals were alive was to understand the relations of structural

and functional relations between organs, *and*, additionally and crucially, the functional interactions between each animal system and its *milieu*. (This is the origin of our meaning of 'environment' as more than just 'surroundings'.)

For Cuvier and his followers, the study of animals was no longer surface-taxonomy plus extrapolations of human physiology. In the new zoology, as in chemistry, the combinations and properties of elements served to explain both structure *and* process, variety *and* its classification (Coleman, 1964; Outram, 1984). Here was the basis of a new 'professional' science for museum curators and university professors. As we shall see below, it was developed most fully in the German universities, and from the later nineteenth century in those of Britain and the USA; it was the mainstay of the zoology curriculum in high schools up to the 1960s. The same kind of analysis was applied to plants, so constituting 'classical botany', a discipline whose trajectory paralleled that of zoology.

In 1800, plant classification had been more developed than that of animals. The comparative anatomy of plants was also better known, in part because 'their anatomy was external' – the main parts (root, stem, leaves, flower parts, etc.) were easily seen in the adult plant and even in the seed. Parisian botanists stressed the different *plans* of plants, notably the deep differences between monocotyledonous plants (such as grasses and palms) and dicotyledonous plants (such as buttercups or oak trees) – the former grew from the base and the leaves had parallel veins, the latter grew from the tips and the leaf veins branched. This division was not to be read as simply an accumulation of differences, but rather as a matter of fundamentally different '*organisation*'. It was akin to Cuvier's focus on structure and function, but the emphasis was on structure – partly because plant physiology was obscure. It was in Paris that a Swiss botanist, Auguste-Pyrame de Candolle, worked out a classification which has remained as the basis of plant taxonomy. It was harder to use than that of Linnaeus, but seemed more natural; professionals liked it, amateurs differed, because they preferred a practical scheme over analytical niceties (Daniels, 1967). In some ways de Candolle

and the German professors of botany were followers of Cuvier, but they also drew on 'morphology', that analysis of form which Cuvier had come to reject (Daudin, 1926; Morton, 1981; Sachs, 1890).

Analysing form

In principle there are at least two ways of analysing structures. So far in this chapter we have discussed the multiple-element-compound model – chemicals, medical bodies, animals and plants are made up of *various* elements, tissues or organs, compounded in different ways with different outcomes. That was the French mode of analysis – it called for 'dissection'. But there was another mode, often followed by those for whom dissection was 'murder' – a mode seen as German and as opposed to the analytical spirit of Paris. It involved the perception of a structural *idea* which 'unfolded' through development; the complex organism, in its whole and its parts, expressed this idea. The best known advocate of this view was Goethe, the German polymath and poet, who saw leaves as the fundamental organs of plants. (The parts of flowers were modified leaves – so, too, were stems and roots.) A plant was not primarily a compound of elements but a transformation and multiplication of a simple element (Cunningham and Jardine, 1990; Lenoir, 1982; Russell, 1916).

That this position and Cuvier's were not entirely opposed may be evident to us, but antagonism grew as morphologists in Paris, notably Cuvier's colleague Etienne Geoffroy Saint-Hilaire, argued for fundamental similarity not just within each of Cuvier's four main kinds of animals but across them. Geoffroy saw arthropods as animals with skeletons on the outside, as vertebrates inside out; Cuvier was appalled and there was a famous showdown at the Paris Academy in 1830, a year of political revolution in France. Cuvier won the debate 'on points', but died soon afterwards; Geoffroy, as champion of the Romantic left enjoyed a burst of popularity. In France, as in Britain, morphology appealed to political radicals, who often linked it with Lamarck's version of transformism to provide a materialist

explanation of the variety and 'adaptedness' of plants and animals – an alternative to the natural theology of the clerics.

Morphology has been overshadowed by 'evolution' but it was then a highly productive science, asking questions about form scarcely thought of before. For example, morphologists were concerned with symmetries: why did the front and back limbs of vertebrates have the same basic structure? Was the skull a series of vertebrae, much modified? These questions remained fertile for much of the nineteenth century, they were often linked to questions about development and embryology, and later they became part of 'evolutionary theory'. But they were initially and logically quite separate from evolution as a historical process. Indeed, much of what is *now* understood as evidence for evolution was first discovered as evidence for the morphologists' 'common vertebrate plan'. As an example we can cite the homologies between ear-ossicles and gill arches, i.e. the demonstrable continuity of form and position across all the intermediaries between the tiny ear-bones of mammals and the bones which support the gills in fish (Appel, 1987).

The study of animal form was also pursued at the microscopic level, and was linked with the medical microscopy mentioned above. The 'general anatomy' of Bichat used tissues as elements, but many anatomists, especially in the German universities began to look for common 'generative' tissues from which the other tissues could be derived. Here again was the Romantic emphasis on unity and common ancestry; here, too, was an incentive to microscopy. Could one find a microstructure common to different tissues? Candidates for that role included the idea that all tissues were made of uniform globules of protein, but this view collapsed when better microscopes were developed in the 1830s (Pickstone, 1973). A new candidate emerged in 1839 from the joint researches of a botanist, Mathias Schleiden, and a young anatomist/physiologist, Theodor Schwann. They argued not so much for a common *unit*, as for a common mode of development: that in all plants and animals, including man, the fluids of the body 'crystallised' to form 'nuclei' around which proteinaceous materials were deposited. Plant cells acquired

walls and fluid vacuoles appeared in the cell-stuff; animal cells did not get walls, and they adopted a wide variety of shapes – from flat skin-cells to elongated muscle fibres. Here was another huge analytical programme, called histology, to excite anyone who could master the use of a microscope: all the tissues of all the plants and animals in the world – adult or embryonic, healthy or diseased – could be analysed in terms of the development of cells (Ackerknecht, 1953; Bynum, 1994; Harris, 1998).

Readers who know modern biology may have spotted a catch. We do not believe that cells are formed out of fluids, any more than we believe that microbes can be so formed; for us cells come from cells, and microbes from microbes. In so knowing we follow the French and German investigators who revised cell theory, not least by examining simple plants, and who argued that microbes were tiny plants produced by other tiny plants. And if we combine these two new perspectives, we see one of the great generalisations of the mid-nineteenth century (and a new meaning of eggs) – all animals and plants were made up of cells, derived from other cells and ultimately from an egg-cell. That egg-cell came from the mother and in sexual reproduction had been fertilised by a sperm-cell (or pollen-cell) from the ‘father’. Now follow back the ancestry – individuals came from parents, cells from cells, as far back as the species went. That grand analytical conception is so familiar to us that we forget it is only 150 years old.

We have already suggested that morphology in general, including embryology and histology, drew much of its strength from the concern of German culture with *development*. Through cell-theory it became closely entwined with studies of plant and animal organisation, and with questions of evolution, as we will briefly explore in the next section.

Classical zoology and botany

What did zoologists and botanists *do* in the nineteenth century? From many histories, including much of the best recent history of science, one would guess that they collected and classified as

part of natural history, that a few of them pioneered experimentalism in biology. and that their overwhelming concern was with evolution, especially debates about Darwinism. This is a distortion, arising in large measure from the 'peculiarities of the English' and the Anglo-American orientation of much of the historical writing. Before about 1870, Britain had very few professional biologists, but it was a world leader for imperial exploration, amateur natural history and natural theology – thus it produced Darwin. The impact of the *Origin of Species*, when published in 1859, was enormous; it dominated later nineteenth-century discussions of natural theology and the relations of science and morals. In our day, Darwin's disciples have made evolutionary theory and sociobiology into lively sciences and major components of 'public understanding' of science. But as Peter Bowler has shown repeatedly, writing against the tide of the 'Darwin Industry', if you read what professional biologists were publishing between 1860 and 1940, Darwinism is much less prominent than one might suppose (Bowler, e.g. 1983; 1988).⁴ I think we can illuminate the situation by remembering that most of the biological sciences were then not just natural history, nor experimentalist; they were analytical. (I have already outlined some of them, and more will follow.)

Consider the problems that Darwin faced as a naturalist who also knew the work of French and German zoologists. Some were geographical – how could you analyse the patterns of animal and plant distribution in terms of climate etc, and in terms of the territories of similar species (Browne, 1983; Nicolson, 1987)? Some were morphological – why were the basic structures of large groups of animals so similar when their modes of life were so different? Why were embryos of different kinds of animals more alike than were the adults? Some were taxonomic – why did some groups of animals and plants seem much more varied than others, at the level of major groupings *and* at the level of subspecies? Could species of animals and plants be represented as 'budding off' from previous species? Such questions were *not* matters of simple observation. They arose from the biologists who had analysed animal forms, shown

how embryos diverged, analysed patterns of classification, and tried to work out laws for the distribution of animal species in space and time. We may see Darwin as pulling together all these forms of analysis and showing how their results could be explained if, indeed, animal species were derived from each other over long periods of time. He was by no means the first to think of that, but he made the case very systematically, using all the best science of his time. And he provided a possible mechanism, natural selection, which many professionals thought insufficient but which most recognised as serious possibility (see Bowler, 1989a; Browne, 1995; Desmond and Moore, 1992).

In this perspective, evolution was to the analytical sciences then sometimes called biology, as energy was to the analytical sciences sometimes called physics. The new doctrines were ways of pulling together previously separate domains, of finding a deeper common ground. In a way that was new for biology, if not for the more mathematical sciences, they were achievements of 'theory', co-ordinating and reorienting existing analytical fields. In an 'age of capital', as the political historian Eric Hobsbawm has suggested, these doctrines were impressive 'accumulations', built on the creations of the 'age of revolution' (Hobsbawm, 1962; 1975). Indeed, the doctrine of evolution, when combined with the 'revised' cell-theory discussed above, meant that all animals and plants could be seen as parts of an enormous lineage of cells, now extending back in time beyond the origin of the species concerned, back to the common ancestor, back to the origin or origins of life. The study of that great genealogy was 'biology', a potential intellectual synthesis of many disciplines held together by these new generalisations.

But just as physicists went on analysing the behaviour of light and heat, electricity, magnetism, etc., so zoologists and botanists went on analysing form and development, usually now at the microscopic level. They went on trying to relate embryology to taxonomy and to palaeontology, as they had done since the early century. They used their museums for comparative studies, and the laboratories for dissections and microscopical studies of structure and development (Bowler, 1996). They worked, for the

most part, as analysts – and some of them, as we shall see in the next chapter, also came to incorporate the methods of experimentalism.

It is easy now to overlook the continuing significance of analytical STM, partly because of the success of ‘experimentalists’ from the later nineteenth century. When ‘experiment’ was proclaimed as essential to science, analysis could seem old-fashioned, as we shall see in Chapter 6. The analysts rarely asserted themselves against the experimentalists’ simple contrast of experiment with ‘mere description’; so it came to be taken for granted. Nowadays, if asked about modern molecular biology, for example, most participants and commentators, even specialist historians, would probably describe an *experimental* science, in contrast to the old kind of biology which was natural history – mere description and classification. But that opposition is naive; to my mind it leaves little room for most of the science of the nineteenth century and much of that which followed. Physiological chemistry, say, or classical embryology are much more than ‘mere description’; the double-helix structure of DNA was revealed, not by experiment as such, but by the systematic mobilisation of a range of *analytical techniques* – notably organic analysis of the bases and X-ray crystallography of suitable crystals. Watson and Crick were successful because they realised the importance of the problem, and because they were well placed to collect the relevant analytical results and to see how they could be fitted together (Abir-Am, 1997; Olby, 1974; Watson, 1968); they did few, if any, ‘experiments’.

Sciences of the earth

Eighteenth-century ‘geology’ had comprised special crafts plus parts of natural history and natural philosophy. Minerals, soils, etc. were described and classified, and stories were offered about earth histories. The former enterprise was continuous with the natural history of plants, the latter with general theories about the nature of the cosmos. There was no separate discipline of

'geology'. That discipline was created as teachers of mining and mining surveyors came to envisage the rocky mantle of the earth as a series of different layers or strata. These were not all to be found at any given site, and sometimes they appeared in the 'wrong order', but a right order could be discovered – an ideal series of strata in terms of which the strata of any given site could be understood. That hypothesis proved productive, strata proved to be reasonably distinct – they did not generally grade into each other, they could be recognised by their mineral compositions (and later, especially, by their *fossils*).

Stratigraphy was a boom science of the early nineteenth century, giving intellectual interest to the appreciation of landscape and promising to be useful for mining industries. Hunters of fossils could classify them by an extension of Cuvier's zoology, or use them to characterise strata (and then to wonder about the succession of life forms). Thus stratigraphy led back into new questions about earth history, and thence into the discussions of evolution which featured so largely in later nineteenth-century biology (and in most historical writings on geology and biology). But as a form of scientific work, as subsidised by governments to facilitate mining, geology was stratigraphical puzzle-solving. Imperial interests extended that mapping project worldwide (Laudan, 1987; 1990; Rudwick, 1972; 1976; 1985; Secord, 1986).

Mineralogy was another new earth science; it came in two kinds according to the elements used. Chemical mineralogy dissected minerals in terms of the chemical elements; crystallography was a kind of morphology, based on the discovery of 'unit crystals' – solid geometrical forms characteristic of particular minerals. Again, the Paris Museum of Natural History was a key site (Laudan, 1987; Metzger, 1918), and again there were questions about the power of professor-curators to take the specimens apart – some of the mineral specimens were 'jewels'.

But stratigraphy and mineralogy were not the only forms of earth analysis that became part of exploration and imperialism from the early nineteenth century. The patterns of the earth's magnetic fields were mapped, and so were isotherms – lines of

equal temperature; all of which enterprises involved the collection of data from many points on the globe. The chief instigator of several such projects was the cosmopolitan German, Alexander von Humboldt (whose brother Wilhelm was one of the chief authors of German university reform). Alexander was well connected in all the scientific capitals of Europe and he travelled widely in America. He was one of the people who mapped patterns of vegetation, in relation to climate, taking account of altitude as well as latitude. It was Humboldt who called for systematic observation of geomagnetism, and Britain was one of the countries to respond. Under the control of the British army, observation stations were established in Montreal, Tasmania, the Cape of Good Hope and Bombay (Bowler, 1992; Browne, 1983; Cannon, 1978; Nicolson, 1987).

That the British navy was heavily engaged in mapping coastal waters throughout the world, is part of the background to the Charles Darwin story. HMS *Beagle*, on which he voyaged in the early 1830s, was a naval ship; the captain, Fitzroy, was an authority on barometric measurements, and there was a naturalist paid to take the trip. (Darwin was the unofficial naturalist, helping reduce the social isolation of the captain for the years away from civilisation.)

As the English natural philosopher John Herschel noted *c.*1830, all these extensions of the world of knowledge added to the richness and to the speed of scientific development which seemed part of that analytical age (Herschel, 1835: ch. 6). And the speed and richness increased further as steamships and telegraphy improved communications. Analytical meteorology became much easier, and more useful, when widespread data on weather, especially barometric pressures, could be collected simultaneously. With such data, the elements of the science as we know it were worked out – the cyclones and the anticyclones that we see each day on the weather forecasts.

In the twentieth century, ‘aerology’ – the study of the atmosphere as a complex dynamic system was pioneered by a Norwegian physicist equipped with sophisticated mathematics (Bowler, 1992). At about the same time, the Manchester physi-

cist Patrick Blackett was collecting data on the geomagnetism of rocks in the ocean floor, so lending support to the once heretical theory of continental drift, and helping initiate the 'tectonic revolution' in earth sciences (Hallam, 1973). He was also encouraging Bernard Lovell to develop cosmic ray studies by constructing a huge radio telescope at Jodrell Bank (Agar, 1998; Lovell, 1976). All these projects were primarily analytical, rather than experimental, indeed they contained much that might be called natural history; they involved observatories and field measurements more than laboratories. They illustrate my general argument that new analytical forms of science have continued to develop and will continue to do so – in academic settings, in the field and 'at work'. But now, again, we return to 1800, to explore the origins of the social sciences.

Analysing the social

One of Michel Foucault's strongest claims about human knowledge was that the human sciences were invented about 1800, and might now be on the way out (Foucault, 1970: ch. 10). I would argue that new social sciences were indeed created, in analytical mode, alongside the new natural sciences. They too were created 'out of' natural-historical knowledges, which were displaced rather than replaced; the natural history continued to matter, even when the new analytical forms were hegemonic.

Foucault showed that eighteenth-century discourses about wealth and language, like those about animals and plants, were mostly of the kind I call natural history; they were classificatory, focused on systems of correspondences, easily expressed in *tableaux*. By contrast, the new sciences of political economy and philology were about 'organisms' – economies or languages with characteristic structures and patterns of change. But were these new sciences analytic in using 'elements', and might the new patterns of work have anything in common with those we have discussed for natural sciences?

Whereas eighteenth-century discourses on wealth were mostly

descriptive or prescriptive, the political economy established in Britain *c.*1820 by the author and financier, David Ricardo, was analytical in its reduction to such key notions as agricultural rents and the productivity of land, or the relations of supply, demand and mechanisation. It is no accident that here we first find the simple diagrams of idealised relations that continue to decorate economics' texts and which look so much like the diagrams in texts of physical chemistry. Analytical economics, henceforth, would be substantially deductive; 'realistic' accounts of business development or national economies have continued to be closer to natural history than to analytic formalisms. This tension is recurrent; one sees it in the late nineteenth-century disputes between 'English' analytic accounts and the sociological studies of the German school; or in British universities now, where prestige economics is mostly mathematical, and less abstract forms are mostly classed as 'development' or 'management science' (Roll, 1973).

Whether historical sociology can help explain the emergence of analytical economics is an open question. In as much as it was first a product of the Scottish universities, it shared a pedagogical context with the emergent medical and chemical sciences. But the chief characteristic of early nineteenth-century political economy in Britain would seem to be its very high public profile and politicisation; Ricardo's formulae, and the related demography of T. R. Malthus, were issues of the day. Ricardo's works on agriculture, subsistence, and the problems of pauperism, like his work on 'the machinery question', were central to political discussion. He taught a nation the principles by which new problems might be managed to the satisfaction of the propertied classes. For this, the authority of 'science' mattered; it was bolstered by the formalism, by the mobilisation of statistics, and by this double similarity to the new physical sciences with which it was closely associated.

Several well-known authors wrote on chemistry or geology, as well as on political economy; the same publishers, journals and institutions 'carried' all these sciences; all taught respect for the 'laws of nature and society', both to the bourgeoisie who bene-

fited thereby and to the working classes who did not (Berg, 1980). As Foucault stressed, the political economy of the early nineteenth century was doubly related to the emergent sciences of life: through the demography of Malthus that focused on human reproduction, and through the questions of agricultural productivity that were central to Ricardian economics (Young, 1985). French 'sociology', as invented by Saint-Simon and Comte was closely related to 'biology' through models of organisation (Haines, 1978; Pickstone, 1981). That the emergent science of philology was also so associated may further the general argument for simultaneous reshaping across the range of STM.

The study of language had been focused on relations between words; it came to explore the differing structures of languages and the rules of their historical transformations: 'the isolation of the Indo-European languages, the constitution of a comparative grammar, the study of inflection, the formulation of the laws of vowel gradation and consonantal changes – in short the whole body of philological work accomplished by Grimm, Schlegel, Rask and Bopp' (Foucault, 1970: 281). Comparative grammar was explicitly analogous to comparative anatomy; and the study of language groups was part of the new study of human variation that also encompassed physical anthropology and aspects of the new archaeology.

There appears to be little work on the social and material aspects of this birth of philology, perhaps, because we easily assume that such science can be done anywhere. But if we examine the best English history of philology, one guided by Kuhnian paradigms rather than by Foucault or materialist sociology, then we notice, more than did the author, the significance of libraries, and especially of the rare collections of exotic manuscripts, e.g. in Sanskrit or Norse, that were investigated by the new philologists (Aarsleff, 1983). Previous writers on language had relied largely on the skills and literary knowledge familiar to most scholars, but the new philology was based on collections that could be investigated in much the same way as collections of pressed plants or animal skeletons. Characteristically, in the

new or reformed universities of Germany, institutes created for the study of philology had *research libraries* and *seminars* for the discussion of research results – two of the tools of scholarship to which we are now almost blind – two keys, perhaps, to the wide extension of analysis across the disciplines. As other historians have pointed out quite independently, the tradition of ‘research’ which was to make German universities famous, began in philology and in mathematics, rather than in the physical sciences (Turner, 1971).

There is room for much extension of the argument. For example, we might regard mathematical statistics as the new form of analysis of numerical data, using means and standard deviations, etc. as ways of conveying the essential characteristics of the aggregate (Hacking, 1990). Demographers, in turn, found statistical ways of characterising populations and their development. And some of the nineteenth-century’s new ‘subjects’ could be regarded as analytical ‘compounds’. For example, the new study of public health was primarily about characterising places – not by natural history or topography, but by drawing on analytical chemistry and stratigraphy as well as on mortality statistics (Hamlin, 1998; Pickstone, 1992b). Phrenology, the science of reading human propensities from the shape of the skull, was the new analytical science of character, based on the location of particular faculties in particular parts of the brain (Cooter, 1984). Nineteenth-century periodicals were full of analytical discourses which we could see as running in parallel and overlapping – e.g. political economy, public health and geology (Rudwick, 1980). Which is not to say that natural history, topography, physiognomy, case histories or simple sets of numbers were excluded; these natural historical forms continued, but in new relations. As we have seen for analytical political economy versus the historical school, the relationships were contested.

By the end of the nineteenth century, even in Britain, ‘social sciences’ were established in key universities. The neo-classical, marginalist economics of W. S. Jevons, Alfred Marshall, etc. was developed in universities – in Manchester, in Cambridge, and

from the early twentieth century in the London School of Economics (LSE). We can treat neo-classicism, with its emphasis on markets as the means of determining prices, as a new form of analysis. Like its classical parent, it was related to contemporary natural sciences – here to kinetic theories in physics (Mirowski, 1989; Roll, 1973; Schabas, 1990). But the decades around 1900 also saw the development of higher education for business managers; this was much less analytical and mathematical – it comprised economic history and the methods of accountancy; after the First World War it could include the physiology of fatigue, much researched among the workers in munitions factories.

We see similar tensions in several countries since the Second World War as government funding for a huge expansion of universities and healthy demand for ‘non-science’ graduates, allowed substantial expansion of the social sciences. For the first quarter-century post-war, much of the emphasis was on the analytical aspects of social sciences – on kinship and structural-functionalism in anthropology, on social structure and mobility in sociology, and on formal linguistics and formalist accounts of literature, etc. The last quarter of the century saw two opposite pulls. Scholars in university departments developed more hermeneutic approaches in anthropology, sociology and cultural studies – the so-called linguistic turn, but among the growing ranks of ‘management scientists’, there was a renewal of more historical, descriptive and indeed prescriptive studies that we might call natural history – loosely linked with such weakly analytical forms of social sciences as occupational psychology and ‘theories of the firm’.

But, of course, ‘natural historical’ and hermeneutic accounts of social phenomena were not, and are not, the prerogative of academics or indeed of ‘investigators’. In social investigations, the ‘objects’ can speak. They have their own systems of meaning, their own ‘natural histories’ and, in some cases, their own modes of analysis. I will touch on these issues in the final chapter, but first I want to pull together some of the issues about ‘analysis’ raised in this chapter and the last, so also preparing for the chapter on experimentalism.

Reflections on the institutions of analysis

Readers who have attended to the various passing remarks about the institutional sites of analysis will have noted that the sections on the decades around 1800 referred frequently to higher professional education and to state museums, hospitals, etc., especially in France, and that later sections referred repeatedly to universities, especially in Germany. From time to time I referred to observatories, surveys, expeditions and field stations. I also stressed the analytical work of ‘consultants’ in industry, agriculture and medicine, the place of analysis and rationalised production in factories, and the roles of analysts in industrial research laboratories. What causal claims, if any are here involved? How am I picturing the relations between these various ‘institutions’ and the ideal-type way of knowing that I call analytical?

Perhaps I need first to say that no account of such complex questions can be both simple and adequate, and that I draw from the best studies of such matters the general lesson that all such causation is reciprocal. People’s ideas obviously influence the institutions they create and the ways the institutions develop but, equally, the institutional patterns in which men and women are raised and work also structure the ways in which they think and act, at a conscious level but also through lots of assumptions and tacit knowledge that the people concerned may never examine. This interplay means that we can reasonably speak of ways of knowing (and of making) as *embodied* in the workings of institutions, where ‘institution’ can cover surveys or factories or, indeed, the working habits of a group of investigators, provided they are sufficiently ingrained and stable to be passed on to any new members of the group. So, for example, if we consider the STM institutions of major European countries c. 1850, I would want to claim that the major teaching hospitals, national museums, state surveys, professional schools and ‘advanced factories’ can sensibly be characterised as chiefly analytical. Staff in such institutions worked on people, specimens, natural features, processes, machines or economies – which they diagnosed, ordered and regulated according to their elements.

Do I want to claim that the formal institutions concerned – e.g. the Museum of Natural History in Paris, or London teaching hospitals or engineering workshops in Manchester necessarily *originated* as analytical institutions? No. It is a historical fact that some of them originated for other reasons and had different ‘modes of operation’. Indeed, in the chapter on natural history I argued that the big state museums grew as ‘treasure houses’ and places of display more than for systematic study, and that where collections were used for systematic study and education, this was mostly as natural history. The same is true of teaching hospitals; in the later eighteenth century ‘walking the wards’ became a prized part of medical education, especially in London, but the cases were ‘displayed’ as individuals; their ‘massing’ in hospitals was a matter of convenience – you could see many individual patients in a short time. We have noted that in some early engineering colleges, education was by a sort of collective apprenticeship, and the layout of many engineering workshops had no obvious rationale in terms of the steps of production. Analysis then loomed small in these collections of specimens, patients, tools and pupils.

But such collections could be (re)made to facilitate analytical enterprises. They could be reordered, materially and/or conceptually, to promote and display analytical knowledges and practices, provided that people who were so directed and motivated had the conceptual and material resources and the social power to effect these transformations and implement new ways of working. By sketching the long history of analysis as an ideal, I have suggested one root for the aspiration. By stressing the discovery of ‘elements’ constitutive of the new fields of knowing, I underlined that aspiration alone will not suffice; creative processes were required that were to some extent unpredictable. But what of the motivations and the power?

For industrial and agricultural analysis the motive seems relatively straightforward – the expectation of material gain by improving productivity through better control of inputs, outputs, processes and by-products. Such expectations may not always have been realised; many commercial processes remained

very resistant to analysis – in which case, craft and lore ('natural history') were better guides. But the expectations were well met in many cases, enough it seems to gradually convince industrialists that analysis was worth paying for, at least when things went wrong. Analysts, of course, actively encouraged such convictions.

For governments, too, economic gain or increased efficiency more generally could be major motives. British politicians of the early nineteenth century were mostly educated in classics rather than natural philosophy, and so were the generals and the admirals, but they recognised a national (and class) interest in the improvement of navigation, agriculture or mining and were persuaded to support data collections and analysis which promised such improvements. When faced with escalating costs and/or unrest over such matters as the relief of the poor and the containment of epidemics, they were willing to lend political clout to the intellectual authority claimed by the economists and their 'reformist' doctor friends. In such ways old institutions were partially converted to analysis, and new institutions, such as the geological survey or the Poor Law Commission, were designed to the plans of analysts (Hamlin, 1998).

In the French professional schools and the German technical schools (as, later in the science faculties of British and American universities), new analytical knowledges were often produced to elucidate the 'principles of practice'. By the end of the century, all major universities and technical colleges had laboratories where chemical apparatus, gas-engines or dynamos were provided so that students could learn the techniques they would later apply 'on site', in 'works' laboratories' or in private 'testing' facilities. All such professional education was closely linked to worlds of practice and, by 1900, to government institutions for regulation and standardisation of practice, such as the National Physical Laboratory (1900) and its German model, the *Physikalische-Technische Reichsanstalt* (Cahan, 1989; Magnello, 2000; Moseley, 1978). This was the institutional network that produced and used analysis. (We shall return to it in Chapter 7; perhaps we could think of it as the analytical version of technoscience.)

Yet I would stress again that natural historical knowledges continued to be important. We have detailed the argument for medicine, but it also applies to other parts of STM. The debates between medical analysts and 'biographical medicine' were paralleled in those between industrial analysts and upholders of craft knowledges. In taxonomy, we have mentioned the tensions between professional botanists much taken with de Candolle's structural classification and those who preferred the easy Linnaean system (Daniels, 1967). Such disputes were common in the nineteenth century, and they remain important. So also for contested boundaries between different kinds of analysts. How, for example, should geological chemistry be related to chemical geology, or how did X-rays compare with tapping and the stethoscope as a way of 'looking into the chest'? Whenever a new analytical technique is invented, the 'classification' which results needs to be squared with existing classifications, and professional 'turf' may be at issue. Moreover, since many analytical techniques came to be practised by technicians rather than the 'principal investigator' or the doctor in charge of the case – so additional boundaries came to be contested.

In stressing analysis as a way of knowing I am trying to highlight features which are common to many fields of STM and over many periods; thus we can facilitate comparative use of data and insights across different cases. Although my chapters trace particular 'ways of knowing' over time, I emphasise again that these are *all* to be understood as *always* developing in dynamic, often contested, relationships – *both between and within the ideal types*.

In the next chapter we turn to experimentalism and invention. These seem to me to have a social dynamic different from analysis: they were not, generally speaking, so closely related to the dissection and regulation of economic and professional practices. Rather, I shall argue, they were developed in 'sanctuaries' deliberately distanced from professional and commercial practices, privileged places where novelties could be created and controlled. These sites were equipped as for analysis, but were associated with incentives to synthesis – rather as early analytical sites had been equipped for natural history but converted to analysis.

Again, it is not hard to see why, in the ever more complex world of nineteenth-century industry, some inventors might try to set up invention-factories, or why some industrialists might relieve their chemists of routine analytical work so as to concentrate on creating new products. But what of the practice of making experiments – how and why were sanctuaries provided that were both academic and ‘constructive’? Part of the answer, especially for France and Britain in the early nineteenth century, lay in a few peculiar institutions such as the Royal Institution and the Collège de France, where professors, often trained in analysis, lectured to fashionable audiences attracted by demonstrations of power over nature. However, the chief answer lay in universities, especially in Germany. By mid-century, partly because of their successes as analysts, many German professors of science had well-equipped laboratories; they also had the time to do as their discipline seemed to require. They did not have to spend all their days on teaching or on analysis of specimens, they could try to build knowledge, so expanding the ‘research ethic’ of the universities. Relying on the training of students for analysis, but not needing to be at the beck and call of practitioners or industrialists, they could create model controlled-worlds, so attracting students and other academics who could be persuaded that such syntheses were a step beyond mere analyses, and that laboratory novelties were more attractive than the messy world outside.

Notes

- 1 For very useful histories of pathological analysis, see Foster (1961; 1983).
- 2 The important works by Jewson (1974; 1976) are often read in this way.
- 3 On germs, see Worboys (2000); on science and management, Sturdy and Cooter (1998); on transformations of British medicine, Lawrence (1994).
- 4 For a discriminating survey, see Bowler (1989a).